

```
In [71]: fruits=['apple','banana','chery','watermilon']
print(fruits)
```

```
['apple', 'banana', 'chery', 'watermilon']
```

```
In [72]: for fruit in fruits:
print(fruit)
```

```
apple
banana
chery
watermilon
```

```
In [73]: for i,fruit in enumerate(fruits):
print(i,fruit)
```

```
0 apple
1 banana
2 chery
3 watermilon
```

```
In [66]: print(fruits[0:3])
```

```
['apple', 'banana', 'chery']
```

```
In [74]: #List method
#append
fruits.append('orange')
print(fruits)
print(len(fruits))
```

```
['apple', 'banana', 'chery', 'watermilon', 'orange']
5
```

```
In [81]: #insert
fruits.insert(1,"kiwi")

print(fruits)
print(len(fruits))
print(fruits[0:4])
```

```
['apple', 'kiwi', 'kiwi', 'kiwi', 'banana']
5
['apple', 'kiwi', 'kiwi', 'kiwi']
```

```
In [76]: #remove
#remove only first occurence
fruits.remove("kiwi")
fruits.remove("orange")
print(fruits)
```

```
['apple', 'banana', 'chery', 'watermilon']
```

```
In [78]: #pop
#remove only last item in list
poped_item=fruits.pop()
print("poped item is:",poped_item)
print(fruits)
```

```
poped item is: chery
['apple', 'banana']
```

```
In [86]: #count
print(fruits)
print(f"How many kiwi in list item:{fruits.count('kiwi')}")
```

```
['apple', 'kiwi', 'kiwi', 'kiwi', 'banana']
How many kiwi in list item:3
```

```
In [89]: #sort
fruits.sort()
print(fruits)
```

```
['apple', 'banana', 'kiwi', 'kiwi', 'kiwi']
```

```
In [94]: #reverse
fruits.reverse()
print(fruits)
```

```
Out[94]: ['kiwi', 'kiwi', 'kiwi', 'banana', 'apple']
```

```
In [97]: #clear
fruits.clear()
print(fruits)
```

```
[]
```

```
In [100... #list slicing
num=[1,2,3,4,5,6,7,8,9,10]
print(num[:])
print(num[2:6])
print(num[6:1:-1])
print(num[7:10])
print(num[:2])
print(num[:4])
print(num[::-1])
```

```
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
[3, 4, 5, 6]
[7, 6, 5, 4, 3]
[8, 9, 10]
[1, 3, 5, 7, 9]
[1, 5, 9]
[10, 9, 8, 7, 6, 5, 4, 3, 2, 1]
```

```
In [104... #list comprehension
list=[]
for i in range(10):
    list.append(i**3) #cube all list element
print(list)
```

```
[0, 1, 8, 27, 64, 125, 216, 343, 512, 729]
```

```
In [106... [i**2 for i in range(10)] #list comprehension
```

```
Out[106... [0, 1, 4, 9, 16, 25, 36, 49, 64, 81]
```

```
In [108... [i**2 for i in range(10) if i%2==0]
```

```
Out[108... [0, 4, 16, 36, 64]
```

```
In [114... #using list comprehension
even_num=[num for num in range(1,100) if num%2==0]
print("Even number between 1 to 100:",even_num)
#using Loop
list=[]
for num in range(1,100):
    if num%2==0:
        list.append(num)
print(list)
```

```
Even number between 1 to 100: [2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98]
[2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98]
```

```
In [117... #nested list comprehensive
list1=[1,2,3,4]
list2=['a','b','c']
pair=[(i,j) for i in list1 for j in list2]
print(pair)
```

```
[(1, 'a'), (1, 'b'), (1, 'c'), (2, 'a'), (2, 'b'), (2, 'c'), (3, 'a'), (3, 'b'), (3, 'c'), (4, 'a'), (4, 'b'), (4, 'c')]
```

```
In [123... words=['Hello','nice','jayanta','']
len_words=[len(word) for word in words ]
len_words
```

```
Out[123... [5, 4, 7, 0]
```